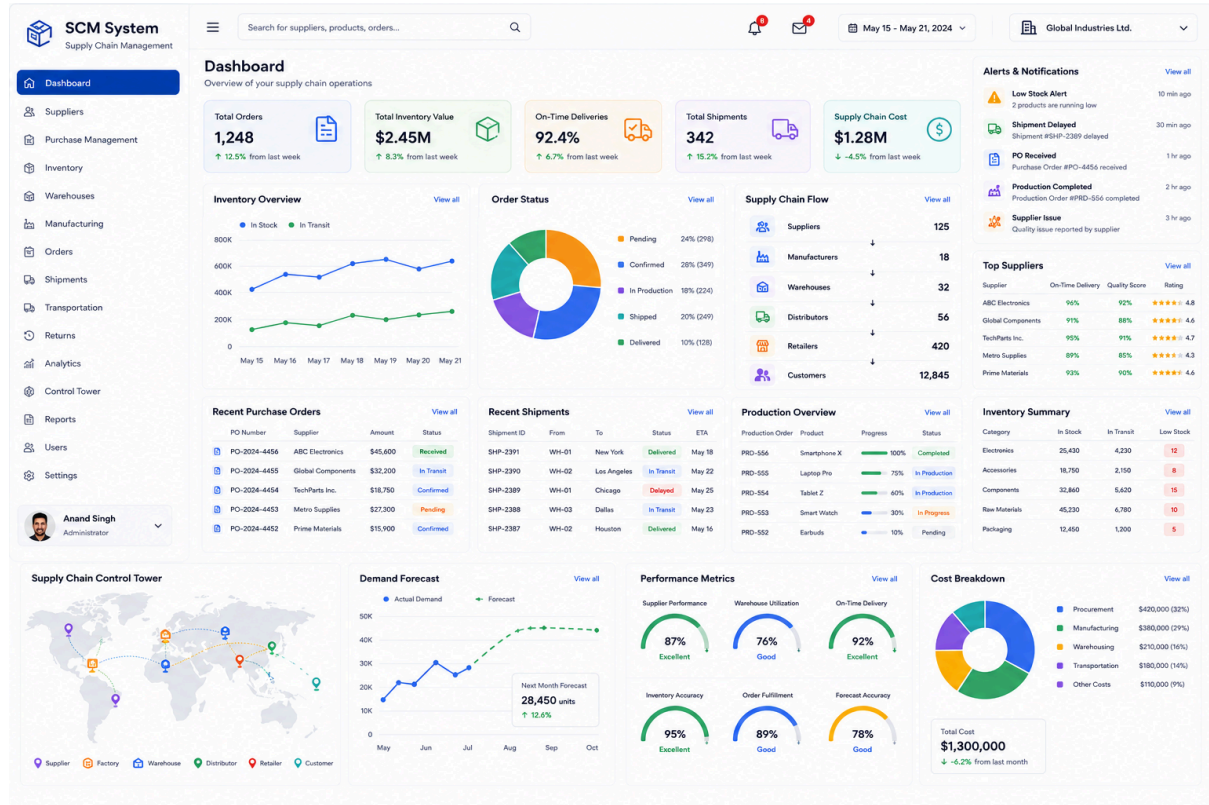


Supply Chain Management System (SCM)



1. Project Introduction

What is a Supply Chain Management System?

A Supply Chain Management (SCM) System is an enterprise software platform that manages the complete flow of products, materials, information, and finances from suppliers to manufacturers, warehouses, distributors, retailers, and finally customers.

A supply chain is much larger than a warehouse.

It manages:

Raw Material Procurement



Supplier Management



Manufacturing



Inventory Management



Warehouse Management



Distribution



Retailers



Customers

This project simulates how large organizations like:

- Amazon
- Flipkart
- Walmart
- Unilever
- Nestlé

manage millions of products across their supply chain.

2. Business Problem

Modern businesses face several supply chain challenges.

Supplier Problems

- Delayed deliveries
 - Poor communication
 - Quality issues
-

Manufacturing Problems

- Raw material shortages
 - Production delays
-

Inventory Problems

- Overstocking
 - Understocking
 - Inventory mismatch
-

Distribution Problems

- Shipment delays
 - Route inefficiencies
-

Business Problems

- Increased operational cost
 - Poor forecasting
 - Reduced customer satisfaction
-

The SCM system solves these issues through centralized supply chain visibility and automation.

3. Project Objective

The platform should allow:

Suppliers

- Supply raw materials
 - Manage purchase orders
-

Manufacturers

- Manage production
 - Track materials
-

Warehouses

- Store inventory
 - Manage stock movement
-

Distributors

- Manage shipments
- Deliver products

Retailers

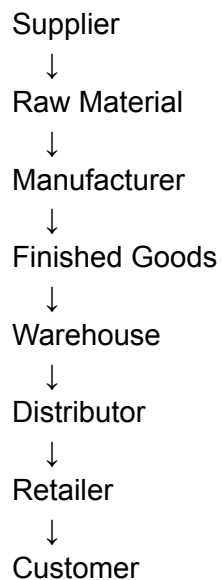
- Place replenishment orders
-

Admin

- Monitor complete supply chain
-

4. Real World Business Workflow

Complete Supply Chain Workflow



Procurement Workflow



Order Fulfillment Workflow

Retailer Order



Inventory Verification



Warehouse Allocation



Shipment Planning



Dispatch



Delivery

5. User Roles

Supplier

Provides raw materials.

Responsibilities:

- Manage inventory
 - Accept purchase orders
 - Deliver materials
-

Manufacturer

Produces goods.

Responsibilities:

- Manage production
 - Track raw materials
-

Warehouse Manager

Stores products.

Responsibilities:

- Manage inventory
 - Process transfers
-

Distributor

Handles logistics.

Responsibilities:

- Manage shipments
 - Deliver products
-

Retailer

Receives products.

Responsibilities:

- Place orders
 - Track deliveries
-

Admin

System administrator.

Responsibilities:

- Monitor complete supply chain
 - Manage stakeholders
-

6. Core Functional Modules

Module 1: Authentication & User Management

Purpose:

Provide secure platform access.

Features

- Registration
 - Login
 - Password Reset
 - JWT Authentication
 - Role-Based Access Control
-

User Information

Name

Organization

Role

Email

Phone

Module 2: Supplier Management

Purpose:

Manage supplier ecosystem.

Supplier Information

Supplier Name

Supplier Code

Products

Location

Contact Person

Performance Score

Features

- Add Supplier
- Supplier Evaluation
- Supplier Ratings
- Supplier Contracts

Example

ABC Electronics

Supplier Score: 92%

Module 3: Product Management

Purpose:

Manage products across supply chain.

Product Information

Product Name

SKU

Category

Cost Price

Selling Price

Unit

Categories

Electronics

Fashion

Food

Pharmaceutical

Furniture

Module 4: Purchase Management

Purpose:

Manage procurement.

Purchase Workflow

Purchase Request



Purchase Order



Supplier Confirmation



Delivery

Purchase Information

PO Number

Supplier

Products

Quantity

Expected Date

Module 5: Inventory Management

One of the most important modules.

Purpose:

Track inventory across the supply chain.

Inventory Information

SKU

Available Stock

Reserved Stock

In Transit Stock

Warehouse Location

Features

- Stock Tracking
 - Stock Transfer
 - Inventory Audit
-

Example

Laptop

Available: 500

In Transit: 150

Module 6: Warehouse Management

Purpose:

Manage warehouse operations.

Warehouse Information

Warehouse Name

Capacity

Location

Manager

Features

- Warehouse Allocation
 - Inventory Storage
 - Capacity Monitoring
-

Module 7: Manufacturing Management

Purpose:

Track production.

Production Workflow

Raw Materials



Production Order



Manufacturing



Finished Goods

Production Information

Production Batch

Raw Materials

Quantity

Status

Features

- Production Scheduling
 - Material Consumption
 - Production Reports
-

Module 8: Order Management

Purpose:

Manage retailer orders.

Order Information

Order Number

Retailer

Products

Quantity

Status

Order Workflow

Order Created



Approved



Allocated



Dispatched



Delivered

Module 9: Distribution Management

Purpose:

Manage product movement.

Features

Shipment Planning

Route Management

Carrier Allocation

Tracking

Distribution Workflow

Warehouse



Shipment



Distributor



Retailer

Module 10: Logistics & Transportation

Purpose:

Manage deliveries.

Transportation Information

Vehicle

Driver

Route

Distance

ETA

Features

- Fleet Management
 - Route Optimization
 - Delivery Scheduling
-

Module 11: Shipment Tracking

Purpose:

Provide visibility.

Shipment Status

Created

Packed

Dispatched

In Transit

Delivered

Tracking Features

Live Tracking

ETA

Delivery Confirmation

Module 12: Demand Forecasting

Purpose:

Predict future demand.

Forecast Factors

Historical Sales

Seasonality

Market Trends

Inventory Levels

Example

Predicted Demand

5000 Units Next Month

Module 13: Returns Management

Purpose:

Handle reverse logistics.

Return Types

Damaged Product

Expired Product

Wrong Product

Customer Return

Workflow

Return Request



Inspection



Restock / Disposal

Module 14: Vendor Performance Management

Purpose:

Measure supplier performance.

KPIs

On-Time Delivery

Quality Score

Response Time

Cost Efficiency

Example

Supplier Score

92%

Module 15: Analytics Dashboard

Procurement Analytics

Purchase Orders

Supplier Performance

Procurement Cost

Inventory Analytics

Stock Levels

Inventory Value

Low Stock Items

Manufacturing Analytics

Production Output

Material Usage

Production Efficiency

Logistics Analytics

Shipment Status

Delivery Time

Transportation Cost

Business Analytics

Revenue

Supply Chain Cost

Profitability

Demand Trends

Module 16: Notification System

Purpose:

Provide operational alerts.

Notifications

Low Stock Alert

Shipment Delayed

Supplier Delivery Received

Purchase Order Approved

Production Completed

Module 17: Supply Chain Control Tower

This is the most advanced module.

Purpose:

Provide complete supply chain visibility.

Dashboard Displays

Suppliers

Factories

Warehouses

Shipments

Retailers

Inventory

Example

Warehouse A

Inventory: 82%

Shipment Delayed: 2

Orders Pending: 45

7. Frontend Screen Planning

Supplier Screens

Dashboard

Purchase Orders

Products

Deliveries

Performance

Manufacturer Screens

Dashboard

Production

Raw Materials

Inventory

Reports

Warehouse Screens

Dashboard

Inventory

Transfers

Shipments

Analytics

Distributor Screens

Dashboard

Deliveries

Vehicles

Tracking

Retailer Screens

Dashboard

Orders

Inventory

Deliveries

Admin Screens

Dashboard

Users

Suppliers

Warehouses

Manufacturing

Reports

Control Tower

8. Database Design

Core Collections

users

suppliers

products

purchaseOrders

inventory

warehouses

productionOrders

shipments

vehicles

retailers

orders

returns

notifications

analytics

9. Business Relationships

Supplier

|
|---- Purchase Orders

Manufacturer

|
|---- Production Orders

Warehouse

|
|---- Inventory

Retailer

|
|---- Orders

Order

|
|---- Shipment

Shipment

|
|---- Tracking

10. Third-Party Integrations

Google Maps API

Purpose:

- Route Optimization
- Shipment Tracking

Socket.IO

Purpose:

- Real-Time Inventory Updates
- Shipment Tracking

Cloudinary

Purpose:

- Documents
- Product Images

Firebase

Purpose:

- Push Notifications

Twilio

Purpose:

- SMS Alerts

QR Code / Barcode System

Purpose:

- Inventory Scanning
- Shipment Verification

PDF Generator

Purpose:

- Purchase Orders
 - Shipment Documents
 - Invoices
-

11. AI Features (Advanced Version)

AI Demand Forecasting

Predict future sales and inventory requirements.

AI Supplier Selection

Recommend best suppliers.

AI Inventory Optimization

Prevent overstocking and stockouts.

AI Route Optimization

Find fastest delivery routes.

AI Risk Prediction

Detect supply chain disruptions.

AI Procurement Recommendations

Suggest purchase quantities.

AI Supply Chain Control Tower

Provide real-time operational insights.

12. MERN Architecture

Frontend (React)

Pages

Dashboard

Procurement

Inventory

Manufacturing

Warehouses

Shipments

Analytics

Control Tower

Libraries

React.js

Redux Toolkit

React Router

Material UI

Axios

Chart.js

Backend (Node.js + Express.js)

Responsibilities

Authentication

Inventory Management

Procurement

Production Tracking

Shipment Tracking

Analytics

Notifications

Major APIs

POST /api/auth/login

POST /api/suppliers

GET /api/inventory

POST /api/purchase-orders

GET /api/shipments

POST /api/production-orders

GET /api/analytics

GET /api/control-tower

MongoDB Collections

users

suppliers

inventory

warehouses

shipments

orders

productionOrders

purchaseOrders

Unique Selling Point (USP)

The biggest differentiator of this project is the **Supply Chain Control Tower**.

It provides a single dashboard that shows:

Supplier Status

Inventory Levels

Production Progress

Warehouse Utilization

Shipment Tracking

Retail Demand

Forecasting Insights

in real time.

Expected Outcome

The Supply Chain Management System should function as a complete enterprise-level supply chain ecosystem where:

- Suppliers manage procurement.
- Manufacturers manage production.
- Warehouses manage inventory.
- Logistics teams manage shipments.
- Retailers manage replenishment.
- Management gains end-to-end visibility across the entire supply chain.

This project introduces students to procurement systems, inventory management, warehouse operations, manufacturing workflows, logistics management, demand forecasting, analytics dashboards, real-time tracking, and enterprise-scale MERN architecture, making it one of the most industry-relevant ERP and enterprise software projects.